3.19 definition: surjective A function $T\colon V \to W$ is called surjective if its range equals W.

To illustrate the definition above, note that of the ranges we computed in 3.17, only the differentiation map is surjective (except that the zero map is surjective in the special case $W = \{0\}$).

Some people use the term onto, which means the same as surjective.

Whether a linear map is surjective de- pends on what we are thinking of as the vector space into which it maps.

3.20 example: surjectivity depends on the target space The differentiation map $D \in \mathcal{L}(\mathcal{P}_5(\mathbf{R}))$ defined by $Dp = p'$ is not surjective, because the polynomial x^5 is not in the range of D. However, the differentiation map $S \in \mathcal{L}(\mathcal{P}_5(\mathbf{R}), \mathcal{P}_4(\mathbf{R}))$ defined by $Sp = p'$ is surjective, because its range equals $\mathcal{P}_4(\mathbf{R})$, which is the vector space into which S maps.

Fundamental Theorem of Linear Maps The next result is so important that it gets a dramatic name.

3.21 fundamental theorem of linear maps Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then range T is finite- dimensional and

$$dimV = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Proof Let $u_1, \ldots, u_m$ be a basis of $\operatorname{null} T$; thus $\dim \operatorname{null} T = m$. The linearly independent list $u_1, \ldots, u_m$ can be extended to a basis

$$u_1, \ldots, u_m, v_1, \ldots, v_n$$

of V (by 2.32). Thus $\dim V = m + n$. To complete the proof, we need to show that range T is finite-dimensional and $\dim \operatorname{range} T = n$. We will do this by proving that $Tv_1, \ldots, Tv_n$ is a basis of range T.

Let $v \in V$. Because $u_1, \ldots, u_m, v_1, \ldots, v_n$ spans V, we can write

$$v = a_1 u_1 + \cdots + a_m u_m + b_1 v_1 + \cdots + b_n v_n,$$

where the a's and b's are in $\mathbf{F}$. Applying T to both sides of this equation, we get

$$Tv = b_1 Tv_1 + \cdots + b_n Tv_n,$$

where the terms of the form Tu_k disappeared because each u_k is in $\operatorname{null} T$. The last equation implies that the list $Tv_1, \ldots, Tv_n$ spans range T. In particular, range T is finite-dimensional.

Section 3B Null Spaces and Ranges 63 To show $Tv_1, \ldots, Tv_n$ is linearly independent, suppose $c_1, \ldots, c_n \in \mathbf{F}$ and $c_1Tv_1 + \cdots + c_nTv_n = 0$. Then $T(c_1v_1 + \cdots + c_nv_n) = 0$. Hence $c_1v_1 + \cdots + c_nv_n \in \operatorname{null} T$. Because $u_1, \ldots, u_m$ spans null T, we can write $c_1v_1 + \cdots + c_nv_n = d_1u_1 + \cdots + d_mu_m$, where the d's are in $\mathbf{F}$. This equation implies that all the c's (and d's) are 0 (because $u_1, \ldots, u_m, v_1, \ldots, v_n$ is linearly independent). Thus $Tv_1, \ldots, Tv_n$ is linearly independent and hence is a basis of range T, as desired.

Now we can show that no linear map from a finite-dimensional vector space to a "smaller" vector space can be injective, where "smaller" is measured by dimension.

3.22 linear map to a lower-dimensional space is not injective Suppose V and W are finite-dimensional vector spaces such that $\dim V > \dim W$. Then no linear map from V to W is injective.

Proof. Let $T \in \mathcal{L}(V, W)$. Then $\dim \operatorname{null} T = \dim V - \dim \operatorname{range} T \geq \dim V - \dim W > 0$, where the first line above comes from the fundamental theorem of linear maps (3.21) and the second line follows from 2.37. The inequality above states that $\dim \operatorname{null} T > 0$. This means that null T contains vectors other than 0. Thus T is not injective (by 3.15). $\square$

3.23 example: linear map from $\mathbf{F}^4$ to $\mathbf{F}^3$ is not injective Define a linear map $T\colon \mathbf{F}^4 \to \mathbf{F}^3$ by $T(z_1, z_2, z_3, z_4) = (\sqrt{7}z_1 + \pi z_2 + z_4, 97z_1 + 3z_2 + 2z_3, z_2 + 6z_3 + 7z_4)$. Because $\dim \mathbf{F}^4 > \dim \mathbf{F}^3$, we can use 3.22 to assert that T is not injective, without doing any calculations.

The next result shows that no linear map from a finite-dimensional vector space to a "bigger" vector space can be surjective, where "bigger" is measured by dimension.

3.24 linear map to a higher-dimensional space is not surjective Suppose V and W are finite-dimensional vector spaces such that $\dim V < \dim W$. Then no linear map from V to W is surjective.

Proof Let $T \in \mathcal{L}(V, W)$. Then $\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T \leq \dim V < \dim W$, where the equality above comes from the fundamental theorem of linear maps (3.21). The inequality above states that $\dim \operatorname{range} T < \dim W$. This means that range T cannot equal W. Thus T is not surjective.

As we will soon see, 3.22 and 3.24 have important consequences in the theory of linear equations. The idea is to express questions about systems of linear equations in terms of linear maps. Let's begin by rephrasing in terms of linear maps the question of whether a homogeneous system of linear equations has a nonzero solution.

Homogeneous, in this context, means that the constant term on the right side of each equation below is 0.

Fix positive integers m and n, and let $A_{j,k} \in \mathbb{F}$ for $j = 1, \ldots, m$ and $k = 1, \ldots, n$. Consider the homogeneous system of lin-equations

$$\sum_{k=1}^{n} A_{1,k} x_k = 0$$

$$\vdots$$

$$\sum_{k=1}^{n} A_{m,k} x_k = 0.$$

Clearly $x_1 = \cdots = x_n = 0$ is a solution of the system of equations above; the question here is whether any other solutions exist.

Define $T \colon \mathbb{F}^n \to \mathbb{F}^m$ by 3.25 $T(x_1, \ldots, x_n) = (\sum_{k=1}^{n} A_{1,k} x_k, \ldots, \sum_{k=1}^{n} A_{m,k} x_k)$.

The equation $T(x_1, \ldots, x_n) = 0$ (the 0 here is the additive identity in $\mathbb{F}^m$, namely, the list of length m of all 0's) is the same as the homogeneous system of linear equations above.

Thus we want to know if null T is strictly bigger than $\{0\}$, which is equivalent to T not being injective (by 3.15). The next result gives an important condition for ensuring that T is not injective.